

Yu-Yen Chang

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Experienced Data Scientist with a PhD in astrophysics and 15+ years of experience in machine learning, computer vision, statistical modeling, and large-scale data analysis across science and engineering. Experienced in leading end-to-end algorithm development and cross-functional research projects involving data and modeling problems.

Experience

- **Senior Data Scientist / Computational Scientist**, Turion Space, USA 01/2025 - Present
 - Lead the development of machine learning and computer vision solutions for satellite imagery and synthetic data.
 - Develop 10+ advanced algorithms for object detection, tracking, image registration, and parameter estimation.
 - Design multi-modal data pipelines integrating multi-spectral, temporal, and contextual data.
 - Apply ML, physics-informed ML, and numerical optimization to complex space and engineering problems.
 - Drive end-to-end algorithm development and collaborate across research, software, and engineering teams.
- **Associate/Assistant Professor**, Physics, NCHU, Taiwan 02/2020 - 01/2025
 - Led a research group applying ML, statistical modeling, and large-scale data analysis to scientific datasets.
 - Developed cloud-based data analysis workflows and achieved 85%+ precision in galaxy property inference.
 - Analyzed, simulated, and visualized 1M+ observations from 30+ databases for collaborative research projects.
 - Led and mentored 10+ students and postdoctoral researchers in data science and scientific computing.
 - Delivered 10+ lectures and 50+ public presentations; published 40+ papers with 6,000+ citations.
- **Insight Data Science Fellow** (20A.DS.SV), INSIGHT, USA 01/2020 - 03/2020
 - Applied NLP and clustering to identify societal concerns from 100K+ unstructured survey responses.
 - Developed location-specific information retrieval with Elucd, achieving 80%+ performance.
 - Integrated text and geographic information to identify location-specific patterns and customer insights.
- **Postdoctoral Research Fellow**, ASIAA, Taiwan 10/2016 - 01/2020
 - Developed machine learning models using XGBoost and Keras to classify and characterize active galaxies.
 - Built predictive models to identify and characterize different populations of supermassive black holes.
 - Integrated 30+ observational bands and 20+ datasets to analyze galaxy evolution across multiple wavelengths.
- **Postdoctoral Research Fellow**, CEA-Saclay, France 10/2014 - 09/2016
 - Created a catalog for 500K+ distant galaxies with state-of-the-art decomposed models.
 - Evaluated size differences (5σ +) between active and non-active galaxies using image processing.
 - Designed reproducible image-processing and statistical analysis workflows for large astronomical imaging datasets.
- **Postdoctoral/Graduate Research Fellow**, MPA, Germany 09/2010 - 06/2014
 - Generated an 800K+ galaxy catalog by implementing Bayesian and chi-square fitting techniques.
 - Performed 3D reconstruction of galaxy shapes from 2D projected images to model galaxy evolution over 11 Gyr.

Education

- **Ph.D.:** Astronomy, Heidelberg University, Germany 12/2013
- **M.S.:** Physics, National Taiwan University, Taiwan 06/2010
- **M.S.:** Radio Astronomy and Space Science, Chalmers University of Technology, Sweden 01/2009
- **B.S.:** Communication Engineering, National Chiao-Tung University, Taiwan 06/2007

Skills

- Machine Learning: Deep Learning, Computer Vision, Statistical Modeling, Predictive Modeling
- Data Science: Data Pipelines, Distributed Computing, Data Modeling, Cloud Computing, Large-scale Data Analysis
- Modeling: Numerical Optimization, Scientific Computing, Simulation, Bayesian Inference, Physics-informed ML
- Computer Vision: Image Processing, Object Detection, Object Tracking, Image Registration, 3D Reconstruction
- Space & Imaging: Satellite Imagery, Geospatial Data, Multi-spectral Data, Synthetic Data, Time-Series Analysis
- ML Engineering: MLOps, Model Deployment, Model Serving, Docker
- Programming: Python, SQL, C++, C, MATLAB, IDL, Mathematica, Bash, Git, Linux
- ML/Data Tools: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, NumPy, Pandas, SciPy